

INSTRUMENTAL VARIABLE ANALYSIS

PART 5 - OTHER CAUSAL MACHINE LEARNING ANALYSES

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INTRODUCTION

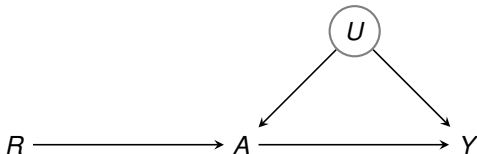
- So far, we have assumed access to **data on all common causes** of treatment and outcome in order to infer a treatment's effect.
- This assumption is **untestable**.
- It may well be violated, especially in the analysis of large databases that were not designed in view of the causal question.
- In that case, progress can sometimes be made using **instrumental variables (IV)**.
- Instrumental variable methods have a long tradition in economics and econometrics.
- They have more recently entered the medical, epidemiological and biostatistical literature.
- The aim is to introduce so-called instrumental variables and clarify how to use them for causal inference.

WHAT IS AN INSTRUMENTAL VARIABLE?

INSTRUMENTAL VARIABLES

instrumental variable (IV): a variable R which

- 1 is associated with the exposure A ;
- 2 has no direct effect on the outcome Y ([exclusion restriction](#));
- 3 does not share common causes with the outcome ([randomization assumption](#)).

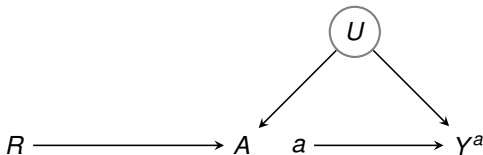


IV ASSUMPTIONS

The exclusion restriction and randomization assumption can equivalently be expressed as

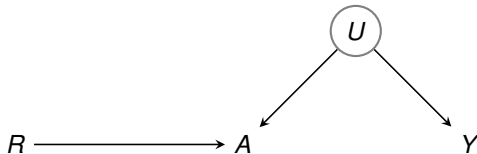
$$Y^a \perp\!\!\!\perp R$$

'no association between outcome and IV in the absence of treatment/exposure'



WHY INSTRUMENTAL VARIABLES?

- IV's can be used for testing whether there is evidence of exposure effects, even in the presence of unmeasured confounders U .
- *How?*

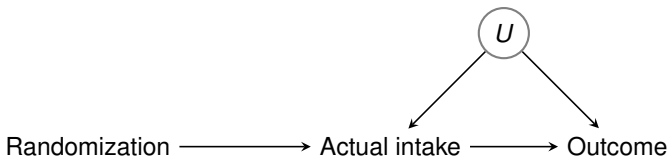


- We will see that they can also be used for effect estimation.

DOUBLE BLIND RANDOMIZED TRIALS WITH NONCOMPLIANCE

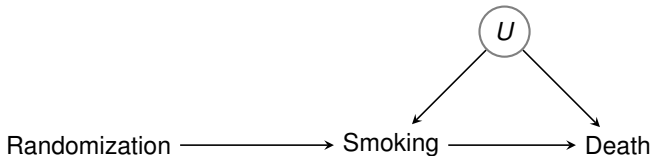
Randomization is an IV for estimating the effect of actual intake on health outcomes because it

- 1 is associated with the exposure;
- 2 has no direct effect on the outcome by blinding;
- 3 does not share common causes with the outcome by randomization.



RANDOMIZED ENCOURAGEMENT DESIGNS

- The Multiple Risk Factors Intervention Trial aims to evaluate the impact of smoking cessation on time to death of myocardial infarction.
- Randomization over encouragement to quit smoking may be an IV for estimating this impact.



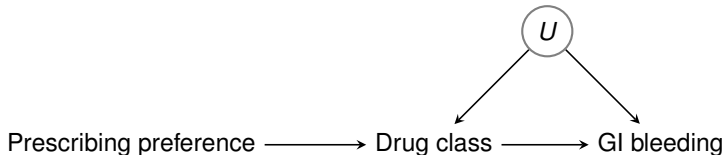
MENDELIAN RANDOMIZATION STUDIES

- The random assortment of genes transferred from parents to offspring - called **Mendelian randomization** - resembles the use of randomization in experiments.
- It is therefore a natural IV for estimating the **effect of genetically affected exposures on a given trait**.
- E.g. the ALDH2 genotype can be used to infer the effect of alcohol consumption as it encodes an enzyme involved in the alcohol metabolism.



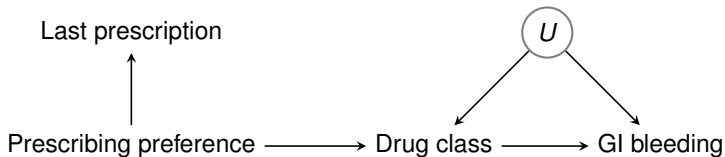
GENERAL INSTRUMENTAL VARIABLES

To compare the effect of prescribing 2 classes of drugs (COX-2 and NSAIDs) on gastrointestinal (GI) bleeding: physician's prescribing preference for drug class as an IV.



GENERAL INSTRUMENTAL VARIABLES

Because physician's prescribing preference for drug class is not directly observed, one may use the physician's last prescription as a surrogate.



HOW TO USE INSTRUMENTAL VARIABLES?

PRINCIPLE

- Assume that the exposure effect can be described using the following model:

$$E(Y|A, R, U) = \alpha + \psi A + U$$

- Then it follows that

$$\begin{aligned} E(Y|R) &= \alpha + E(U) + \psi E(A|R) \\ &= \alpha^* + \psi E(A|R) \end{aligned}$$

under the IV assumptions.

- IV estimates for ψ can thus be obtained by least-squares fitting of

$$E(Y|R) = \alpha^* + \psi \underbrace{E(A|R)}_{\text{exposure model}}$$

IN PRACTICE

Two-stage least squares estimation

Perform 2 least squares regressions:

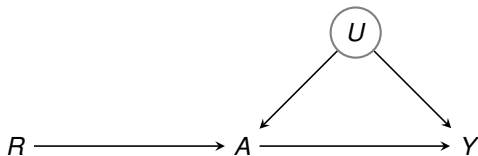
- stage 1: regress exposure on the instrumental variable and obtain predictions $\hat{A}$; e.g.

$$E(A|R) = \gamma_0 + \gamma_1 R$$

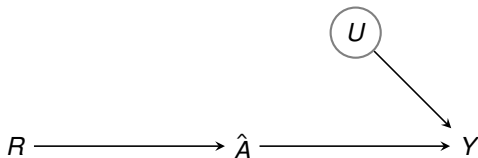
- stage 2: regress outcome on predicted exposure; e.g.

$$E(Y|R) = \alpha + \psi \hat{A}$$

INTUITION



By predicting the exposure on the basis of the instrumental variable (and covariates), we kill its association with the unmeasured confounder U .



CIGARETTE CONSUMPTION

(Hanck et al., 2021)

- How much must taxes be increased to reach a certain reduction in cigarette consumption?
- Economists use **elasticities** to answer this kind of question.
- We use observations on cigarette consumption and several economic indicators for all 48 continental federal states of the U.S. in 1995 to estimate the elasticity ψ in model

$$E(Y|A, R, U) = \alpha + \psi A + U$$

with Y the log average number of cigarette packs per capita sold
and A the log after-tax average real price per pack of cigarettes.

- We use R , the portion of taxes on cigarettes arising from the general sales tax (in dollars per pack) as an instrumental variable.
- It is associated with A and reasonable that it only influences consumption through price.

ANALYSIS IN R

```
> install.packages("AER")
> library(AER)
> library(SuperLearner)
> data("CigarettesSW")

# compute real per capita prices
> CigarettesSW$rprice <- with(CigarettesSW, price / cpi)
> CigarettesSW$rincome <- with(CigarettesSW, income / population / cpi)

# compute the sales tax
> CigarettesSW$salestax <- with(CigarettesSW, (taxs - tax) / cpi)

# generate a subset for the year 1995
c1995 <- subset(CigarettesSW, year == "1995")

# check the correlation between sales tax and price
> cor(c1995$salestax, c1995$rprice)
[1] 0.6856138
```

ANALYSIS IN R

```
> y <- log(c1995$packs)
> r <- c1995$salestax
> a <- log(c1995$rprice)
> moda <- lm(a~r)
> summary(lm(y~fitted(moda)))
```

Residuals:

Min	1Q	Median	3Q	Max
-0.63180	-0.15802	0.00524	0.13574	0.61434

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.7199	1.8012	5.396	2.3e-06 ***
fitted(moda)	-1.0836	0.3766	-2.877	0.00607 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

A 10% price increase reduces the geometric mean consumption by 10% (95% 3% to 16%).

$(\exp(-1.0836 \times \log(1.1)) = 0.90)$

DEBIASED MACHINE LEARNING

COVARIATE ADJUSTMENT

- State income L impacts the demand for cigarettes and correlates with sales tax: states with high personal income tend to generate tax revenues by income taxes and less by sales taxes.
- This may violate the instrumental variables assumptions.
- We will therefore consider adjustment for covariates L , thereby relaxing the IV assumptions to

$$Y \perp\!\!\!\perp R|A, L, U \quad \text{and} \quad U \perp\!\!\!\perp R|L$$

or

$$Y^a \perp\!\!\!\perp R|L \quad \text{for } a = 0, 1$$

- Such adjustment can be done through the first and second stage regressions, but this generates the 'usual' concerns about traditional modeling.
- We will here consider debiased machine learning instead.

A NONPARAMETRIC ESTIMAND

- Consider model

$$E(Y|A, L, R, U) = \psi A + g(L, U)$$

for some unknown function $g(\cdot)$.

- *What does this model assume?*
- The parameter ψ can be defined nonparametrically as

$$\psi = \frac{E \{ \text{Cov}(R, Y|L) \}}{E \{ \text{Cov}(R, A|L) \}}$$

- This identity also holds under **structural mean model**

$$E(Y - Y^0|A, L, R) = \psi A,$$

which has the advantage of not conditioning on unmeasured variables.

WHAT IF THE MODEL IS MISSPECIFIED?

- When

$$E(Y|A, L, R, U) = \psi(L)A + g(L, U)$$

then

$$\psi = \frac{E\{\text{Cov}(R, Y|L)\}}{E\{\text{Cov}(R, A|L)\}} = \frac{E\{w(L)\psi(L)\}}{E\{w(L)\}}$$

reduces to a weighted average of exposure effects, for weights $w(L) = \text{Cov}(R, A|L)$.

- Note that these weights are no longer guaranteed to be positive for all L , which may complicated interpretation.

DEBIASED MACHINE LEARNING

- The influence curve of

$$\psi = \frac{E \{ \text{Cov}(R, Y|L) \}}{E \{ \text{Cov}(R, A|L) \}}$$

equals

$$\frac{\{R - E(R|L)\} [Y - E(Y|L) - \psi \{A - E(A|L)\}]}{E [\{R - E(R|L)\} \{A - E(A|L)\}]}$$

- *How would you use this to construct an estimator of ψ and of its standard error?*
- Note that the resulting estimator may be sensitive to **weak instrument bias**, which occurs when $E \{ \text{Cov}(R, A|L) \}$ is close to 0.

DEBIASED MACHINE LEARNING

- Settings the influence curve's sample average to zero and solving for ψ delivers the closed-form estimator

$$\hat{\psi} = \frac{\sum_{i=1}^n \{R_i - \hat{E}(R_i|L_i)\} \{Y_i - \hat{E}(Y_i|L_i)\}}{\sum_{i=1}^n \{R_i - \hat{E}(R_i|L_i)\} \{A_i - \hat{E}(A_i|L_i)\}}$$

for data-adaptive estimates $\hat{E}(R_i|L_i)$, $\hat{E}(A_i|L_i)$ and $\hat{E}(Y_i|L_i)$.

- Its standard error can be estimated as 1 over root-n times the sample standard deviation of

$$\frac{\{R_i - \hat{E}(R_i|L_i)\} \left[Y_i - \hat{E}(Y_i|L_i) - \hat{\psi} \{A_i - \hat{E}(A_i|L_i)\} \right]}{n^{-1} \sum_{i=1}^n \{R_i - \hat{E}(R_i|L_i)\} \{A_i - \hat{E}(A_i|L_i)\}}$$

- Cross-fitting is recommended.

ANALYSIS IN R

```
> sl.y <- SuperLearner(Y=y,X=data.frame(1),SL.library=c("SL.glm","SL.gam","SL.randomForest"))
> sl.a <- SuperLearner(Y=a,X=data.frame(1),SL.library=c("SL.glm","SL.gam","SL.randomForest"))
> sl.r <- SuperLearner(Y=r,X=data.frame(1),SL.library=c("SL.glm","SL.gam","SL.randomForest"))
> m.y <- predict(sl.y,newdata=data.frame(1))$pred
> m.a <- predict(sl.a,newdata=data.frame(1))$pred
> m.r <- predict(sl.r,newdata=data.frame(1))$pred
> psi <- mean((r-m.r)*(y-m.y))/mean((r-m.r)*(a-m.a))
> psi

[1] -1.011858

> sd((r-m.r)*(y-m.y-psi*(a-m.a)))/(mean((r-m.r)*(a-m.a))*sqrt(length(y)))

[1] 0.3537387
```

A 10% price increase reduces the geometric mean consumption by 9% (95% 3% to 15%).